A Lightweight Attention Mechanism for Low-Resource Text Classification

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Abstract

We introduce SparseGated Attention, a parameter-efficient attention variant that reduces the quadratic cost of self-attention to near-linear while retaining most of its expressive power. On three low-resource classification benchmarks the method matches a full transformer baseline within 0.4 F1 points using 31% of the parameters.

1 Introduction

Transformer models dominate text classification, yet their memory footprint remains a barrier for deployment in low-resource settings. Prior work has explored sparsity, low-rank factorisation, and kernel approximations, each trading accuracy for efficiency.

2 Method

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Given an input sequence of embeddings, we compute a gating scalar per head and apply a top-k sparsification over the attention logits. The gate is trained jointly with the rest of the network using a straight-through estimator.

3 Experiments

We evaluate on three datasets with fewer than 5,000 training examples each. All models are trained for 30 epochs with AdamW and a linear warmup schedule.

4 Results

SparseGated Attention achieves 87.1 average F1 compared to 87.5 for the full baseline, while using 31% of the parameters and 44% of the inference FLOPs.